

Jimin Ryu

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EDUCATION

University of Bristol

Sep 2022 – Jun 2025

BSc Computer Science, First Class Honours

EXPERIENCE

Software Engineer

Jun 2025 – Present

Dassault Systèmes

- Contributed to the migration of a native application to a web-based platform, supporting architectural decisions and ensuring functional consistency across platforms.
- Designed and implemented application features using React and TypeScript, translating native workflows into maintainable web-based solutions.
- Developed and maintained core platform features in C++, implementing new functionality and resolving defects through root-cause analysis.
- Collaborated with backend engineers, designers, and QA to ensure correctness, performance, and alignment with existing systems.

Research Assistant

Jul 2024 – Sep 2024

Hanyang University

- Implemented VR locomotion systems in Unity, integrating vibrotactile feedback and omnidirectional treadmill input for controlled experimental studies.
- Designed and conducted two experimental conditions (static controller-based steering and dynamic walking), managing participant sessions and data collection for 80 subjects.
- Analysed quantitative cybersickness, task load, and presence metrics, collaborating with researchers to interpret results and refine experimental design.

Co-creator (Ca-Pow! Project)

Oct 2023 – Mar 2024

Royal Academy of Engineering

- Co-created and facilitated workshops supporting intersectionally disadvantaged students, tailoring content to address barriers in higher education.
- Led mentor training sessions on inclusive practices, strengthening support structures for underrepresented students.

PUBLICATIONS

Do Vibrotactile-Rendered Virtual Walking Sensations Reduce VR Cybersickness?

- Co-authored a peer-reviewed paper accepted at the **IEEE International Symposium on Mixed and Augmented Reality (ISMAR 2025)**.
- Demonstrated that vibrotactile feedback simulating virtual walking significantly reduced VR cybersickness across static and dynamic locomotion conditions without increasing task load or reducing presence.

PROJECTS

Bridging the Gap: Comparing Geometric Path Planning Between GPT-4o and Classical Algorithms in Indoor Navigation

- Built an experimental pipeline for indoor navigation reasoning using 2D floorplans and Oriented Bounding Box (OBB) data, enabling systematic evaluation of LLM-generated navigation paths.
- Evaluated zero-shot and few-shot prompting for geometric reasoning by defining path-quality metrics and identifying failure modes against classical path-planning baselines.

Haunted Escape — GameJam 2024 Winner (Best Visuals)

- Designed and implemented 15 progressively challenging stages featuring dynamic traps and AI-driven enemy behaviour.
- Recognized for outstanding visual design, atmosphere, lighting, and environmental storytelling.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, Java, C#, Go, SQL

Frameworks/Tools: AWS, React, Unity, Git